CESSDA Metadata Aggregator

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EDDI 2021, Dec 1st, 2021

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What is a Metadata Aggregator?

- A metadata aggregator…
  - … collects metadata from OAI-PMH end points
  - … provides an OAI-PMH end point for re-harvesting the records
  - … **preserves provenance information** in the OAI envelope
  - … is for machine consumption
    - CESSDA Data Catalogue is for humans
- Aggregator does not survive alone, it requires supporting services
  - OAI-PMH Harvester
    - Get records from SP's
  - Metadata Validator
    - Make sure the Aggregator is fed with only record conforming to requirements

Goals, Non-goals & Benefits

- Goals
  - Produce aggregator for CESSDA; to be run in CESSDA environment
  - Allow for larger exposure of metadata without any additional burden to SPs
    - One-stop-shop for machine access to CESSDA Service Providers' metadata
- Non-goals
  - Replace CESSDA SPs OAI-PMH end points
  - Enhance SPs metadata
- Benefits
  - Increase visibility and discoverability of Service Providers' data assets
  - Promote interoperability and metadata compatibility
  - Strengthen cooperation between organisations
    - In and out of CESSDA

Partners

- Co-operation with **CESSDA MO** and **Finnish Social Science Data Archive FSD**
  - FSD doing development
    - Aggregator code bases
  - MO doing integration to CESSDA Cloud
    - Harvesting, validation, platform
    - Mandate
- Build on existing OAI-PMH software
  - Kuha2
  - Open Source
- Timeline
  - Development 2021
  - Production 2022

YHTEISKUNTATIETEELLINEN TIETOARKISTO
FINNISH SOCIAL SCIENCE DATA ARCHIVE

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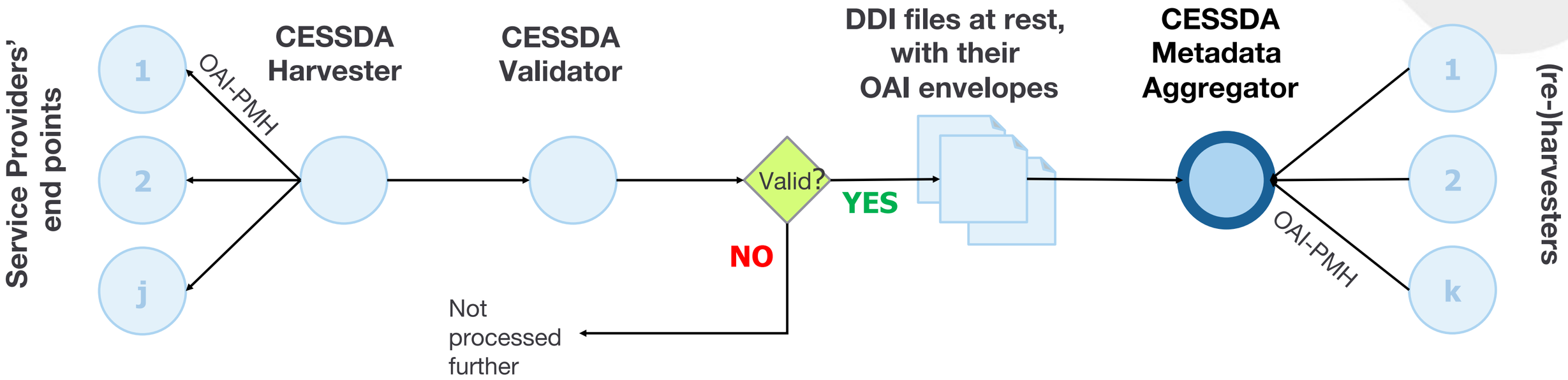
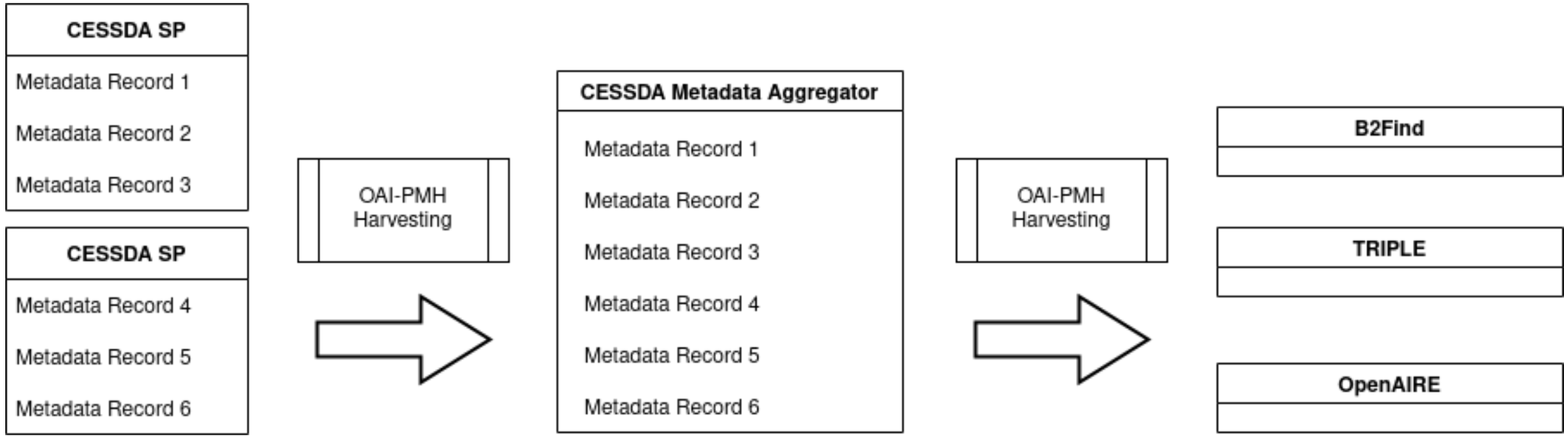
World According to the Aggregator



Software Product

Primary Function

- Gather metadata from CESSDA SP OAI-PMH endpoints
  - Supports DDI 1.2.2, DDI 2.5 and DDI 3.1 as input formats
- Makes the metadata re-harvestable from a single OAI-PMH endpoint
  - Supports OAI-DC, DDI 2.5 and Datacite as output formats



OAI-PMH Features

Common Features

- Supports protocol version 2.0
- Implements all OAI-PMH verbs
- Selective harvesting
  - Sets
  - Datestamps
- Deleted records
- ResumptionTokens
- OAI-Identifiers
  - Configurable namespace identifiers

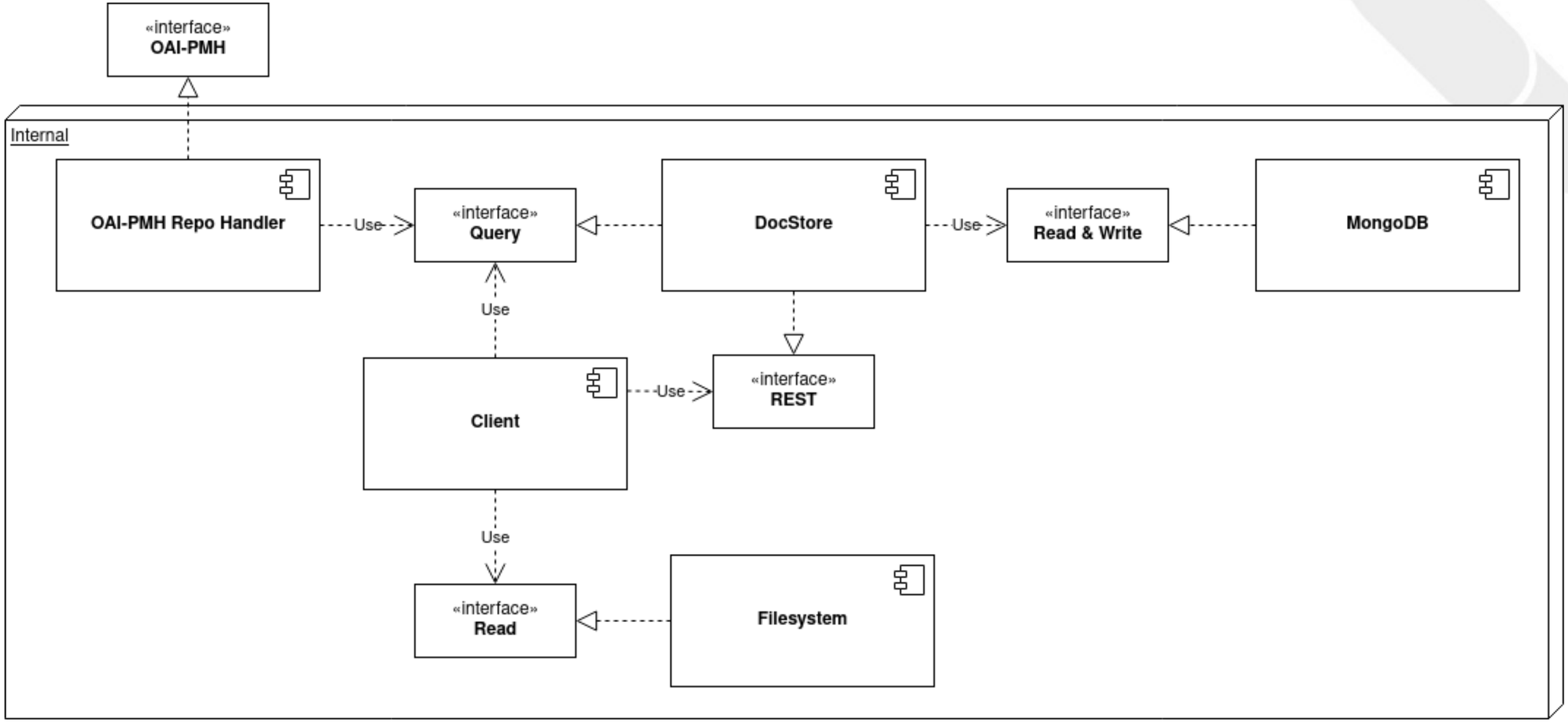
Aggregator Features

- Group records to OAI-sets
  - by source archive
  - by arbitrary criteria
  - using configurable mapping files
- Origin of metadata
  - is compiled from source OAI-PMH envelopes
  - can be traced using provenance containers defined in OAI-PMH specification

Other Features

- Database API – Document Store
  - HTTP server using JSON as communication payload
  - RESTful
  - Rich query support
- Database – MongoDB
  - Support replicas
  - Design is not tied to any metadataformat
- Build new output metadataformats as plugins
- Deduplicate records based on provenance

Components


«interface»
OAI-PMH
Internal
OAI-PMH Repo Handler
Use
«interface»
Query
DocStore
Use
«interface»
Read & Write
MongoDB
Use
Client
Use
«interface»
REST
Use
«interface»
Read
Filesystem

Technical Details

- Software bundle consists of two server applications, a client application and a database
- Applications are written in Python
- Follows CESSDA Technical guidelines
- Uses Kuha2 as a dependency
  - Kuha2 is stable codebase. First release in Q4/2017.
  - Easy to implement new features baked into Kuha2, for example new output and input metadataformats.

Project Status

Immidiate Plans

1 **SHIP IT!**

- Almost done
- Finalise code bases, documentation etc.
  - Target date Dec 31st, 2021

2 **GET IT RUNNING!**

- Running in the production environment
  - Populated and publicly available
- Make it harvestable by 3rd parties
  - Early 2022?

3 PREPARE FOR FUTURE

- Public source code repositories
  - Maintenance
  - Future needs?

Related Efforts Presented at EDDI 2021

Kleemola, M., Maartens, C., Flügel, A. (2021, Dec 1). Harvesting DDI for an integrated catalogue. In Oliver Hopt (Chair), *Interoperability* [Presentation] European DDI User Conference 2021, Paris, France.

Shepherdson, J., Morris, M. (2021, Dec 1). CESSDA's approach to bulk DDI study-level metadata validation and feedback. In Henri Ala-Lahti (Chair), *Workflows* [Presentation] European DDI User Conference 2021, Paris, France.

References based on provisional programme as of Oct 24, 2021.

Thank you! Got questions?

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